



HTML & CSS

Lecture 9





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AOS animation



AOS animation

AOS(Animate On Scroll) is a Java Script library that allows you to create smooth and elegant animations when elements scroll into view on a webpage.

AOS
Animate on Scroll

Installation of AOS

Animate on Scroll

You can also get the AOS link and script to make AOS animation

```
Terminal  
npx install aos
```

You must have node js
to install AOS with npm
keyword

```
index.html  
1 <!DOCTYPE html>  
2 <html lang="en">  
3 <head>  
4   <title>Document</title>  
5   <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/aos@2.3.4/dist/aos.css">  
6 </head>  
7 <body>  
8  
9 </body>  
10  
11 <script src="https://cdn.jsdelivr.net/npm/aos@2.3.4/dist/aos.js"></script>  
12 <script>  
13   AOS.init();  
14 </script>  
15  
16 </html>
```


Using AOS library

Animate on Scroll

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <title>Document</title>
5   <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/aos@2.3.4/dist/aos.css">
6 </head>
7 <body>
8
9   <div data-aos="fade-left"
10     data-aos-anchor="#example-anchor"
11     data-aos-offset="500"
12     data-aos-duration="500">
13   </div>
14
15 </body>
16
17 <script src="https://cdn.jsdelivr.net/npm/aos@2.3.4/dist/aos.js"></script>
18 <script>
19   AOS.init();
20 </script>
21
22 </html>
```



Transition

CSS Transition

Transition is a property that allow you to make a smooth animation from on state to others

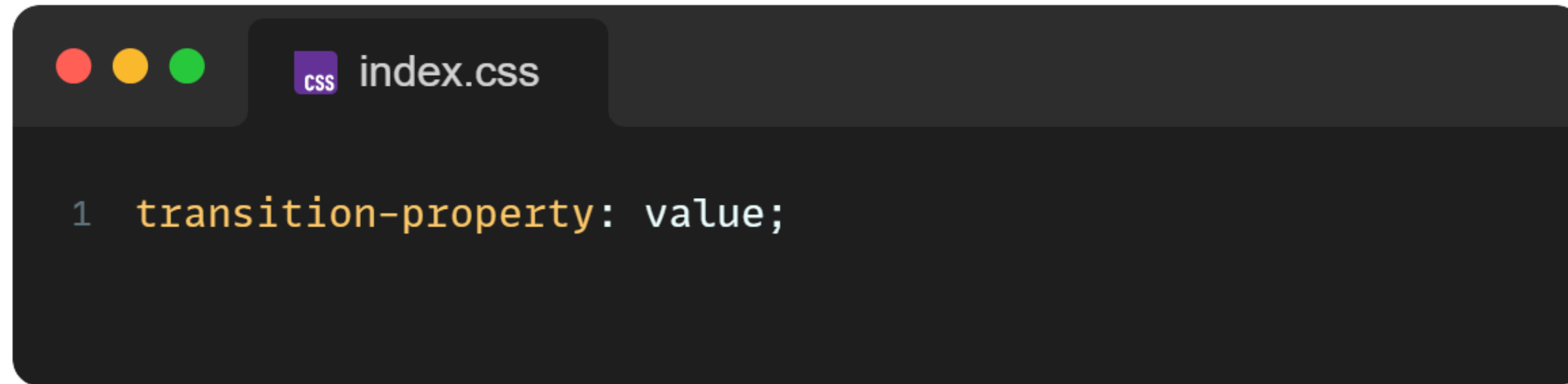
```
index.css
1 button{
2   transition: property duration timing-function delay;
3 }
```

```
index.css
1 button{
2   transition-property
3   transition-timing-function
4   transition-duration
5   transition-delay
6 }
```

We can also give all transition properties in one transition property

Transition property

Syntax:

A code editor window with a dark background. The title bar shows three colored circles (red, yellow, green) and a tab labeled 'index.css' with a 'css' icon. The code content is:

```
1 transition-property: value;
```

```
1 transition-property: value;
```

Specifies which property or properties will be impacted by the transition. It can get 4 type of value.

- all (default)
- single property name
- none
- multiply property names, separated by commas

Transition duration

Syntax:

Specifies the amount of time the transition will take.

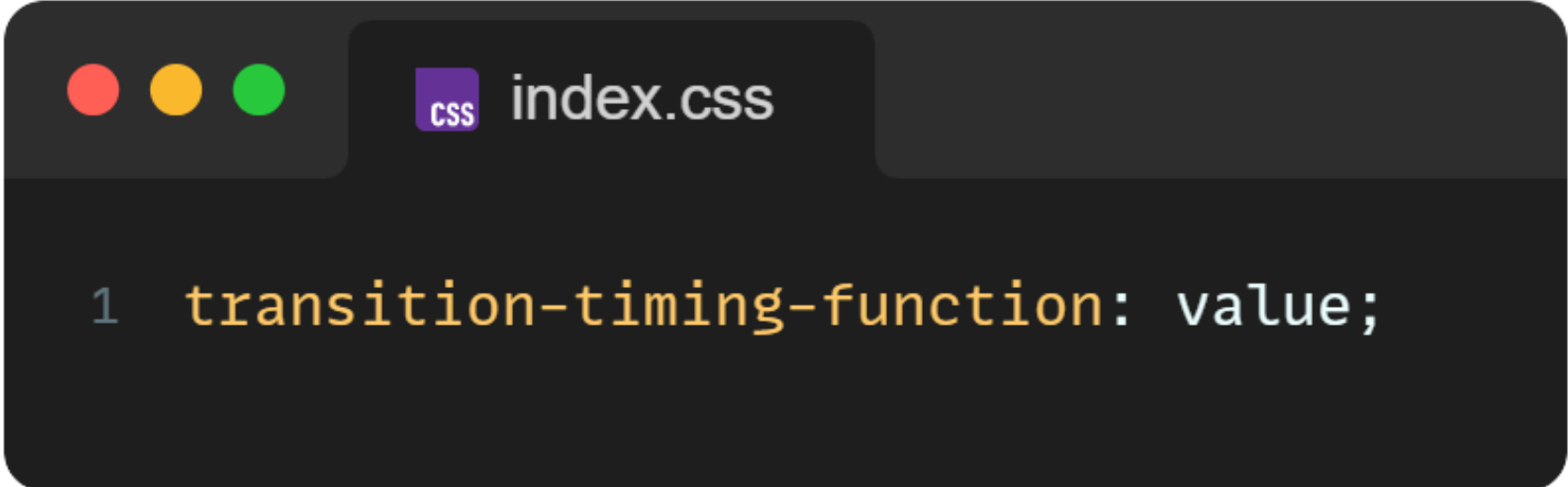
A code editor window with a dark background. The title bar shows three colored circles (red, yellow, green) and a tab labeled 'index.css' with a 'css' icon. The code inside is:

```
1 transition-duration: duration(time);
```

```
1 transition-duration: duration(time);
```

Transition timing-function

Syntax:

A code editor window with a dark background. At the top, there are three colored circles (red, yellow, green) and a tab labeled 'index.css' with a 'css' icon. The main area shows a single line of code: '1 transition-timing-function: value;'.

```
1 transition-timing-function: value;
```

Specifies the pace at which the transition will run.

- **ease** - defines transition with slow start, that fast and end slowly
- **linear** - specifies a transition effect with the same speed from start to end
- **ease-in** - specifies a transition effect with a slow start
- **ease-out** - specifies a transition effect with a slow end
- **ease-in-out** - specifies a transition effect with a slow start and end

Transition delay

Syntax:

Specifies when the transition will start relative to the instant the property value changes.

A code editor window titled 'index.css' with a CSS icon. It contains a single line of CSS code: '1 transition-delay: value;'. The text 'transition-delay' is highlighted in orange, and 'value' is in white. The editor has a dark background and standard window controls (red, yellow, green buttons) on the left.

```
1 transition-delay: value;
```



Kayframes Animation

What is Animation?

- Animation is the art of creating motion or visual effects using CSS.
- It makes webpages interactive, dynamic, and engaging by adding movement, color changes, or transformations to elements.
- The key properties for animation are @keyframes, animation-name, animation-duration, and others that control how and when the animation happens

Kayframes animation

CSS Kayframes animation allow you to create a complex and advanced animation for you webpage

Kayframes defines the specific state of animation where the animation should change

Syntax

A code editor window with a dark background and a tab labeled 'index.css' with a CSS icon. It contains three lines of CSS code: '1 animation: values;', '2', and '3 @keyframes {}'.

```
1 animation: values;  
2  
3 @keyframes {}
```

Animation property

- **animation-name**: Names the animation.
- **animation-duration**: How long the animation lasts.
- **animation-timing-function**: Controls the animation speed curve.
- **animation-delay**: Delays the start of the animation.
- **animation-iteration-count**: Defines how many times the animation runs.
- **animation-direction**: Specifies the animation cycle direction.
- **animation-play-state**: Pauses or resumes the animation.
- **animation**: Shorthand for all animation-related properties

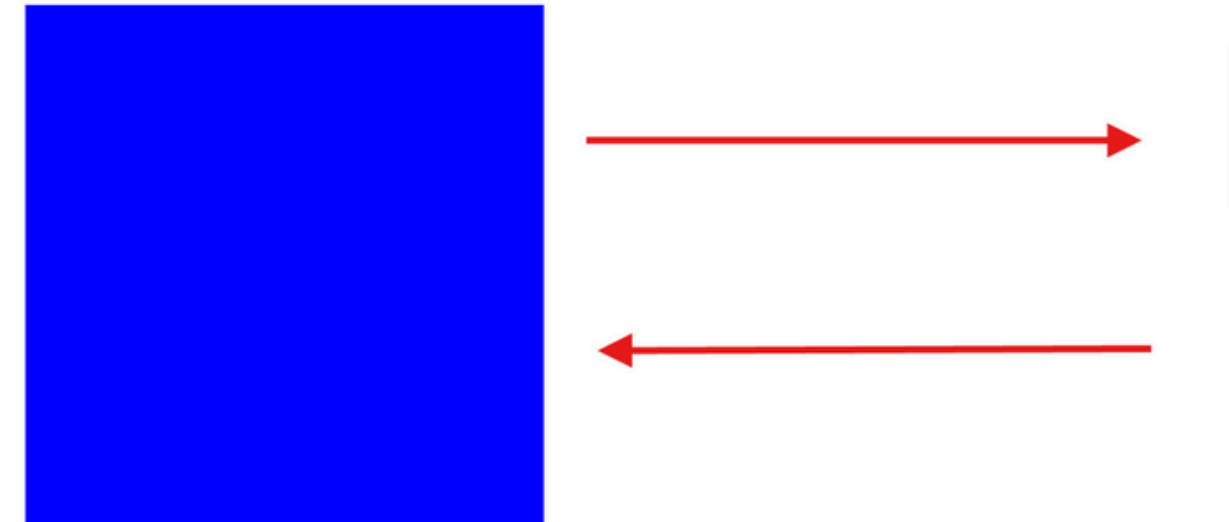
Example Code

Editor



```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <style>
5     div {
6       width: 100px;
7       height: 100px;
8       background-color: blue;
9       animation: moveRight 2s infinite;
10    }
11    @keyframes moveRight {
12      0% {
13        margin-left: 0px;
14      }
15
16      50% {
17        margin-left: 100px;
18      }
19
20      100% {
21        margin-left: 0px;
22      }
23    }
24  </style>
25  <title>Simple Animation</title>
26 </head>
27 <body>
28   <div></div>
29 </body>
30 </html>
```

Result



Browser



THANK YOU!

“If you do what you always did, you will get what you always got.”

A graphic with a dark, moody background of a winding road through a forest. The text 'STAY FOCUSED.' is overlaid in large, white, bold, sans-serif capital letters. A small 'quotefancy' watermark is visible at the bottom center of the image.

STAY
FOCUSED.